

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2018- 2019 (Even Semester)

Innovative Practices Description

UNIT III DC Machines

Degree, Semester& Branch: II Semester B.E. ECE

Course Code & Title: BE8254 Basic Electrical and Instrumentation Engineering

Name of the Faculty member: Mr.G.Sivakumar

Name of the Topic: Construction of DC Motor

Name of the Innovative Practice: Hardware Demonstration

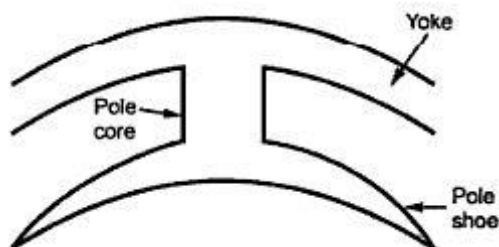
Description:

Poles:

Poles are used to support the field windings. Generally, the field windings are wound on the poles and they are parallel or serially connected with the armature windings. Also, the poles are given joint to the yoke using welding process or by using screws.

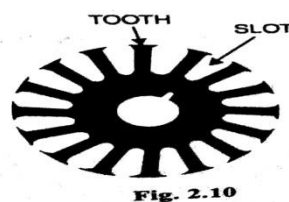
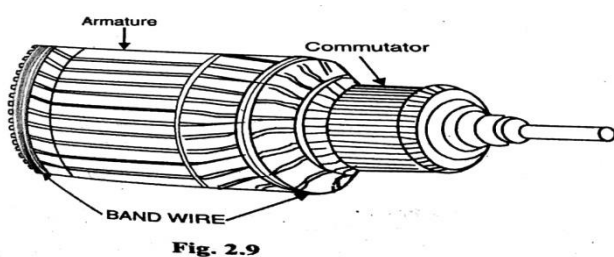
Pole shoe:

The main function of the pole shoe is to spread the magnetic flux and to prevent the field coil from slipping.



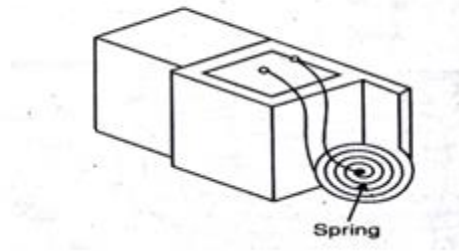
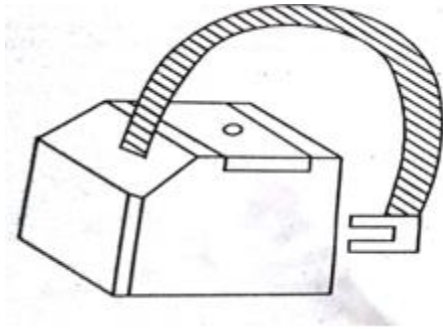
Commutator :

Commutator acts as a rectifier converting AC voltage in the armature winding to the DC voltage across the brushes. It is made of segments of Cu. Each Cu segment is insulated from each other using sheets of mica and is placed on the shaft of the machine.



Brushes:

It ensures electrical connections between the commutator and the external load circuit.



Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	2	2	1	-	-	-	-	1	-	1	-	1

CO3: The students will be able to describe the characteristics of DC Generator and Motor.

References:

1. <https://electricallive.com/2015/03/construction-of-practical-dc-machine.html>
2. Giorgio Rizzoni, "Principles and Applications of Electrical Engineering", McGraw Hill Education (India) Private Limited, 2010.